T3 UV Lagrangian terms grouped by field content

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. The BSM-only UV expression is preferred; the full UV Lagrangian is used as a fallback when the BSM-only conversion is unavailable.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$

$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C^{ij;AB}_{LLS_1S_2}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$

Field combinations 1 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1{}^D$	S_2
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$-S_1{S_1}^\dagger m_1^2$ $+\frac{1}{2}S_1^2\left({S_1}^\dagger\right)^2\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_1\right)^\dagger D_{\mu_1}\left(S_1\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-S_1{S_1}^\dagger m_1^2$ $+\frac{1}{2}S_1^2\left({S_1}^\dagger\right)^2\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_1\right)^\dagger D_{\mu_1}\left(S_1\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-S_1{S_1}^\dagger m_1^2$ $+\frac{1}{2}S_1^2\left({S_1}^\dagger\right)^2\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_1\right)^\dagger D_{\mu_1}\left(S_1\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda_{S_1}^{(2)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_2{S_2}^\dagger$ $+\frac{1}{2}S_2^2\left({S_2}^\dagger\right)^2\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1\,D$	S_2
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda_{S_1}^{(2)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_2 {S_2}^\dagger$ $+\frac{1}{2}S_2^2\left({S_2}^\dagger\right)^2\lambda_{S_2}^{(1)}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda_{S_1}^{(2)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_2 {S_2}^\dagger$ $+\frac{1}{2}S_2^2\left({S_2}^\dagger\right)^2\lambda_{S_2}^{(1)}$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda_{S_1}^{(2)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda_{S_1}^{(2)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda_{S_1}^{(2)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-S_{1,j_1}{S_{1,j_1}}^\dagger m_1^2$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda_{S_1}^{(2)}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-m_2^2 S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}$

Field combinations 2 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2 D	F	F D
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{r_1}}\,F_{r_1}M_F$	$+i\overline{F_{r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{r_1}\right)$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{1}{2}{F_{r_1}}^T\,C\,F_{r_1}M_F$	$+\frac{1}{2}i{F_{r_1}}^T\,C\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{r_1}\right)$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{r_1}}\,F_{r_1}M_F$	$+i\overline{F_{r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{r_1}\right)$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{1}{2}{F_{j_1,r_1}}^T\,C\,F_{j_1,r_1}M_F$	$+\frac{1}{2}i{F_{j_1,r_1}}^T\,C\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}\left(S_2\right)^\dagger D_{\mu_1}\left(S_2\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+D_{\mu_1}\left(S_2\right)^\dagger D_{\mu_1}\left(S_2\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+D_{\mu_1}\left(S_2\right)^\dagger D_{\mu_1}\left(S_2\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\overline{F_{j_1,r_1}}\,F_{j_1,r_1}M_F$	$+i\overline{F_{j_1,r_1}}\,\gamma^{\mu_1}\,D_{\mu_1}\left(F_{j_1,r_1}\right)$

Field combinations 3 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,H$	$S_1\,S_2$	$S_2\,H$
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$+H_{a_2}S_1{H_{a_1}}^{\dagger}{S_1}^{\dagger}{\mathcal{I}_{H1,a_1,a_2}}^{1}\lambda^{(1)}_{HS_1}$	$+S_1S_{2,j_2}{S_1}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{12,j_1,j_2}}^{1}\lambda^{(1)}_{12}$	$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_2}$
cont.										
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+H_{a_2}S_1{H_{a_1}}^{\dagger}{S_1}^{\dagger}{\mathcal{I}_{H1,a_1,a_2}}^{1}\lambda^{(1)}_{HS_1}$	$+S_1S_{2,j_2}{S_1}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{12,j_1,j_2}}^{1}\lambda^{(1)}_{12}$	$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_2}$
cont.										
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+H_{a_2}S_1{H_{a_1}}^{\dagger}{S_1}^{\dagger}{\mathcal{I}_{H1,a_1,a_2}}^{1}\lambda^{(1)}_{HS_1}$	$+S_1S_{2,j_2}{S_1}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{12,j_1,j_2}}^{1}\lambda^{(1)}_{12}$	$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_2}$
cont.										
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{1}\lambda^{(1)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{2}\lambda^{(2)}_{12}$	$+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{1}\lambda^{(1)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{2}\lambda^{(2)}_{12}$	$+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{1}\lambda^{(1)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{2}\lambda^{(2)}_{12}$	$+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{1}\lambda^{(1)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{2}\lambda^{(2)}_{12}$	$+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{1}\lambda^{(1)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{2}\lambda^{(2)}_{12}$	$+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{1}\lambda^{(1)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{2}\lambda^{(2)}_{12}$	$+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_1}$	$+S_2S_{1,j_2}{S_2}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{12,j_1,j_2}}^{1}\lambda^{(1)}_{12}$	$+H_{a_2}S_2{H_{a_1}}^{\dagger}{S_2}^{\dagger}{\mathcal{I}_{H2,a_1,a_2}}^{1}\lambda^{(1)}_{HS_2}$
cont.										
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_2S_{1,j_2}{S_2}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{12,j_1,j_2}}^{1}\lambda^{(1)}_{12}$	$+H_{a_2}S_2{H_{a_1}}^{\dagger}{S_2}^{\dagger}{\mathcal{I}_{H2,a_1,a_2}}^{1}\lambda^{(1)}_{HS_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,H$	$S_1\,S_2$	$S_2\,H$
cont.								$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_1}$		
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_2S_{1,j_2}{S_2}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{12,j_1,j_2}}^{1}\lambda^{(1)}_{12}$	$+H_{a_2}S_2{H_{a_1}}^{\dagger}{S_2}^{\dagger}{\mathcal{I}_{H2,a_1,a_2}}^{1}\lambda^{(1)}_{HS_2}$
cont.								$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_1}$		
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{1}\lambda^{(1)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{2}\lambda^{(2)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{3}\lambda^{(3)}_{12}$	$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$
cont.								$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_1}$		$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{1}\lambda^{(1)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{2}\lambda^{(2)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{3}\lambda^{(3)}_{12}$	$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$
cont.								$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_1}$		$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{1}\lambda^{(1)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{2}\lambda^{(2)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{3}\lambda^{(3)}_{12}$	$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$
cont.								$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_1}$		$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_1}$ $+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_1}$	$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{1}\lambda^{(1)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{2}\lambda^{(2)}_{12}$ $+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^{\dagger}{S_{2,j_2}}^{\dagger}{\mathcal{I}_{12,j_1,j_3,j_2,j_4}}^{3}\lambda^{(3)}_{12}$	$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{1}\lambda^{(1)}_{HS_2}$ $+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_1,j_2}}^{2}\lambda^{(2)}_{HS_2}$
cont.								$+\frac{1}{2}H_{a_2}S_{1,j_2}{H_{a_1}}^{\dagger}{S_{1,j_1}}^{\dagger}{\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_1}$		$+\frac{1}{2}H_{a_2}S_{2,j_2}{H_{a_1}}^{\dagger}{S_{2,j_1}}^{\dagger}{\mathcal{I}_{H2,a_1,a_2,j_2,j_1}}^{2}\lambda^{(2)}_{HS_2}$

Field combinations 4 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,H$	$S_1\,F\,\ell$	$S_2\,F\,\ell$
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$+\overline{\lambda_5} S_{2,j_1} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_1{}^\dagger$ $+H_{a_1} H_{a_2} S_1 \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_1 y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1}} S_1{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1,j_1} {S_{2,j_1}}^\dagger$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\overline{\lambda_5} S_{2,j_1} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_1{}^\dagger$ $+H_{a_1} H_{a_2} S_1 \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_1 y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1}} S_1{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1,j_1} {S_{2,j_1}}^\dagger$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\overline{\lambda_5} S_{2,j_1} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_1{}^\dagger$ $+H_{a_1} H_{a_2} S_1 \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_1 y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1}} S_1{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1,j_1} {S_{2,j_1}}^\dagger$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\overline{\lambda_5} S_{2,j_2} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1} {S_{2,j_1}}^\dagger$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\overline{\lambda_5} S_{2,j_2} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, P_R\, F_{r_1} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {F_{r_1}}^T C\, P_L\, L_{a_1,i_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}} {F_{r_1}}^T C\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1} {S_{2,j_1}}^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\overline{\lambda_5} S_{2,j_2} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1} {S_{2,j_1}}^\dagger$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$+\overline{\lambda_5} S_{2,j_2} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1,j_1} {S_{2,j_1}}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\overline{\lambda_5} S_{2,j_2} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {F_{j_1,r_1}}^T C\, P_L\, L_{a_1,i_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}} {F_{j_1,r_1}}^T C\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1,j_1} {S_{2,j_1}}^\dagger$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\overline{\lambda_5} S_{2,j_2} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1,j_1} {S_{2,j_1}}^\dagger$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+H_{a_1} H_{a_2} S_{1,j_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_5 {S_2}^\dagger$ $+\overline{\lambda_5} S_2 \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_2 \mathcal{C}_{LS_2} {F^*_{a_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1} {S_2}^\dagger$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+H_{a_1} H_{a_2} S_{1,j_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_5 {S_2}^\dagger$ $+\overline{\lambda_5} S_2 \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_2 \mathcal{C}_{LS_2} {F^*_{a_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1} {S_2}^\dagger$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+H_{a_1} H_{a_2} S_{1,j_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_5 {S_2}^\dagger$ $+\overline{\lambda_5} S_2 \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_2 \mathcal{C}_{LS_2} {F^*_{a_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1} {S_2}^\dagger$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$+\overline{\lambda_5} S_{2,j_2} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1,j_1} {S_{2,j_1}}^\dagger$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\overline{\lambda_5} S_{2,j_2} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1,j_1} {S_{2,j_1}}^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\overline{\lambda_5} S_{2,j_2} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1,j_1} {S_{2,j_1}}^\dagger$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\overline{\lambda_5} S_{2,j_2} \mathcal{I}_{LLS_1 S_2}{}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger$	$+\overline{L_{a_1,i_1}}\, C\, P_R\, \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{y_1,a_1,j_1,j_1}$ $+\overline{y_{1,i_1,r_1}} {L_{a_1,i_1}}^T C\, P_L\, F_{j_1,r_1} \mathcal{C}_{LS_1} {F^*_{a_1,j_1,j_1}} S_{1,j_1}{}^\dagger$	$+\overline{y_{2,i_1,r_1}}\overline{F_{j_1,r_1}}\, P_L\, L_{a_1,i_1} S_{2,j_1} \mathcal{C}_{LS_2} {F^*_{a_1,j_1,j_1}}$ $+\overline{L_{a_1,i_1}}\, P_R\, F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_2,a_1,j_1,j_1} {S_{2,j_1}}^\dagger$